

On-Chip Real-Time Polarization Demultiplexing using an Adaptive MZI Photonic Mesh

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Abstract We demonstrate on-chip real-time demultiplexing of two 25 Gbps signals with orthogonal, randomly scrambled polarizations. The photonic circuit consists of a polarization-insensitive 4x2 reconfigurable MZI mesh that also compensates phase drifts of the input signals and is adaptively configured through monolithically integrated electronic multiplexers. ©2026 The Author(s)

Introduction

Coherent optical processors based on Mach-Zehnder interferometer (MZI) meshes enable adaptive manipulation of optical signals for communication and sensing applications [1–4]. In addition to coherent combining, these architectures can recover and separate mixed optical channels by compensating for phase and polarization fluctuations. Recent results have further highlighted the potential of integrated mesh-based circuits for all-optical MIMO demultiplexing in dynamically mixed transmission channels. [5].

In this work, we present a polarization- and phase-insensitive electronic-photonic integrated circuit based on a reconfigurable diagonal MZI mesh that can automatically demultiplex pairs of signals with orthogonal polarization states, even though such polarizations are randomly scrambled. Using integrated monitoring photodiodes and a minimization-based feedback algorithm, the circuit automatically compensates for random phase drifts of the input signals in order to perform coherent recombination and separation of two

orthogonal scrambled polarization states at distinct outputs. This functionality is supported by monolithically integrated electronics implementing time-multiplexed control.

Electronic-Photonic Integrated Circuit Design

Figure 1(a) shows the architecture of the silicon electronic-photonic integrated circuit (EPIC). The chip features four optical inputs provided through edge couplers (ECs) supporting both polarizations. Each EC is connected to a polarization beam rotator splitter (PBRS), which separates and rotates the input polarization components into two output waveguides with identical TE polarization. The resulting eight waveguides are routed to the inputs of a mesh of balanced Mach-Zehnder interferometers (MZIs), while the two mesh outputs are connected to grating couplers (GCs) for vertical fiber coupling off-chip (OUT1 & OUT2).

The mesh consists of an 8×2 array of balanced MZIs arranged along two diagonals. Each MZI building block, highlighted in yellow, comprises two 50:50 directional couplers (DCs) and is thermally actuated by two micro-heaters: one placed

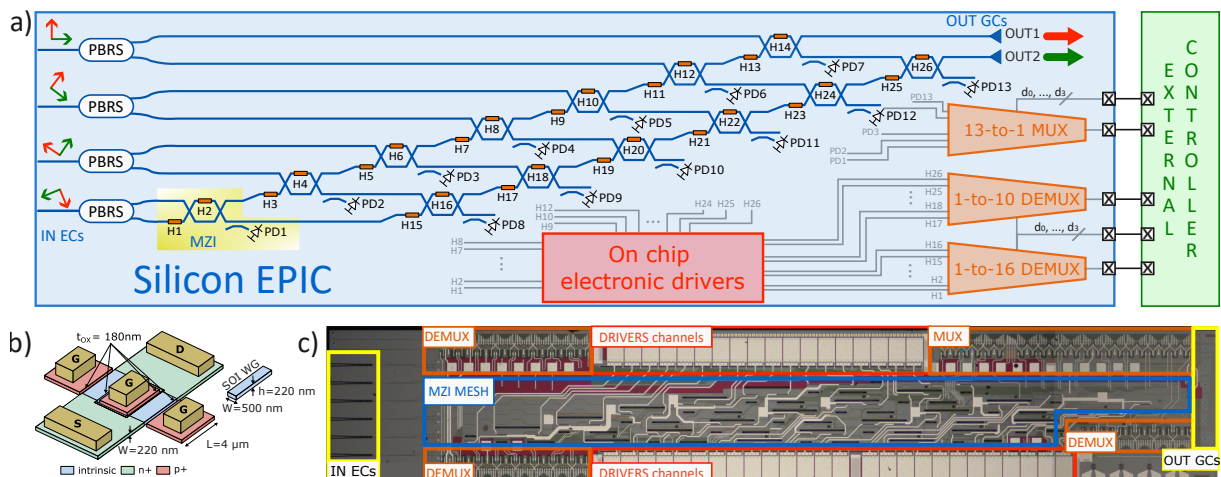


Fig. 1: a) Schematic of the Silicon EPIC, consisting of a dual-diagonal MZI mesh fed by PBRS and controlled by on-chip multiplexing circuitry and external controller, b) Schematic view of the lateral MOS transistor, c) Photograph of the fabricated chip (overall footprint of $6.3 \text{ mm} \times 1.1 \text{ mm}$), highlighting MZI Mesh, integrated electronic drivers, multiplexers (MUX/DEMUX), input ECs, and output GCs. The additional MZI diagonals on the chip are not used.

before the first DC and the other located in the inner arm between the two DCs, providing two degrees of freedom for tuning a MZI. The bottom output of each MZI is tapped via a 90:10 DC and routed to an integrated germanium photodiode (PD), enabling monitoring and stabilization of the MZI operating point.

The MZIs are sequentially tuned by minimizing their bottom output, thereby progressively combining the optical power from all mesh inputs into the first output (red-labeled beam). A second, orthogonal input passes through the first diagonal, coupling only to the MZI output ports; by exploiting the second MZI diagonal, its optical power is coherently combined at the second mesh output (green-labeled beam).

The electronic monolithically integrated section enables time-multiplexed control of the 13 balanced MZIs. The MOS transistors used to realize multiple on-chip circuitry are realized using an unconventional lateral topology compatible with the design rules of a commercial silicon photonic foundry (Advanced Micro Foundry, Singapore). Each device incorporates four channels to increase current capability and operates as a fully depleted silicon-on-insulator (SOI) MOS transistor, enabling both n- and p-type implementations through appropriate doping of the contact regions (Fig. 1b)[6].

The readout chain exploits a 13-to-1 on-chip multiplexer (MUX), which sequentially routes the current signals from the monitoring photodiodes (PDs) to a single acquisition channel. The commands are delivered to the photonic circuit through an actuation chain composed of two integrated demultiplexers (1-to-16 and 1-to-10 DEMUX) and dedicated driver stages, which distribute the control voltages to the MZI heaters. The feedback loop is based on a minimization strategy, where the optical power measured at the tapped ports is sequentially reduced, leading to coherent recombination of the input optical fields.

Automatic tuning of each MZI is achieved via a dithering technique, where a small modulation at 1.2 kHz is applied to the heaters to extract gradient information and drive the system toward the optimal operating point.

A photograph of the die is shown in Figure 1(c), highlighting the different sections of the designed chip.

Single MZI Building Block Characterization

The characterization of the fundamental building block of the circuit, namely a single MZI, is first performed. For this purpose, light is injected using a single-mode fiber into the bottom input of the mesh. As illustrated in the schematic of Fig. 2(a), the input light, with an arbitrary polarization state, enters the chip, where its orthogonal components

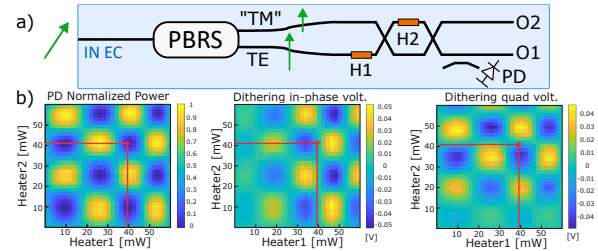


Fig. 2: (a) Schematic of a single MZI used to characterize the mesh building block; (b) 2D maps of the normalized PD power, together with the extracted in-phase and quadrature dithering components, as H1 and H2 are swept.

are separated and rotated into TE polarization by the PBRS. These components are then routed to the MZI inputs.

Figure 2(b) (left) shows the 2D map of the normalized power measured at the integrated PD as the heater voltages H1 and H2 are swept, for a given input polarization state. By selecting a minimum in this map, all the input power is directed to the upper output of the MZI (O2). This condition is achieved by nulling the dithering signal applied to the heaters. The middle and right panels of Fig. 2(b) show the extracted dithering signals from the PD while H1 and H2 are swept. In this case, a single modulation frequency of 1.2 kHz is used, with in-phase and quadrature components applied to the two heaters.

Optical Setup & Data Transmission Experiment

The proposed PIC is used in the optical setup shown in Fig. 3(a) to demonstrate polarization combining and demultiplexing of two input signals. Two 25 Gbps NRZ OOK signals, carried on orthogonal polarization states X and Y, are generated using an external Mach-Zehnder modulator (MZM), with a laser operating at 1550 nm. The modulated signals are subsequently passed through a digital polarization scrambler to control the input state of polarization before entering the chip. The optical signal is then equally divided into four paths using a 1-by-4 fiber splitter (FS), with each path matched in length. These outputs are connected to a fiber array aligned with the four input ECs of the EPIC. At the output of the chip, a fiber collects the stabilized signal from either OUT1 or OUT2 and delivers it to the remaining stages of the setup for modulated signal detection.

First, optical coherent combining is demonstrated using the first diagonal of the proposed EPIC. In this case, ten different polarization states are generated using the polarization scrambler (controller) to vary the input conditions to the mesh. Figure 3(b) shows the normalized power measured at OUT1 for the untuned and tuned cases, for each polarization state. As can be seen, the output power initially fluctuates at OUT1 due to phase variations in the input fibers. After tuning the mesh, that is, after all the MZIs in the first diagonal are

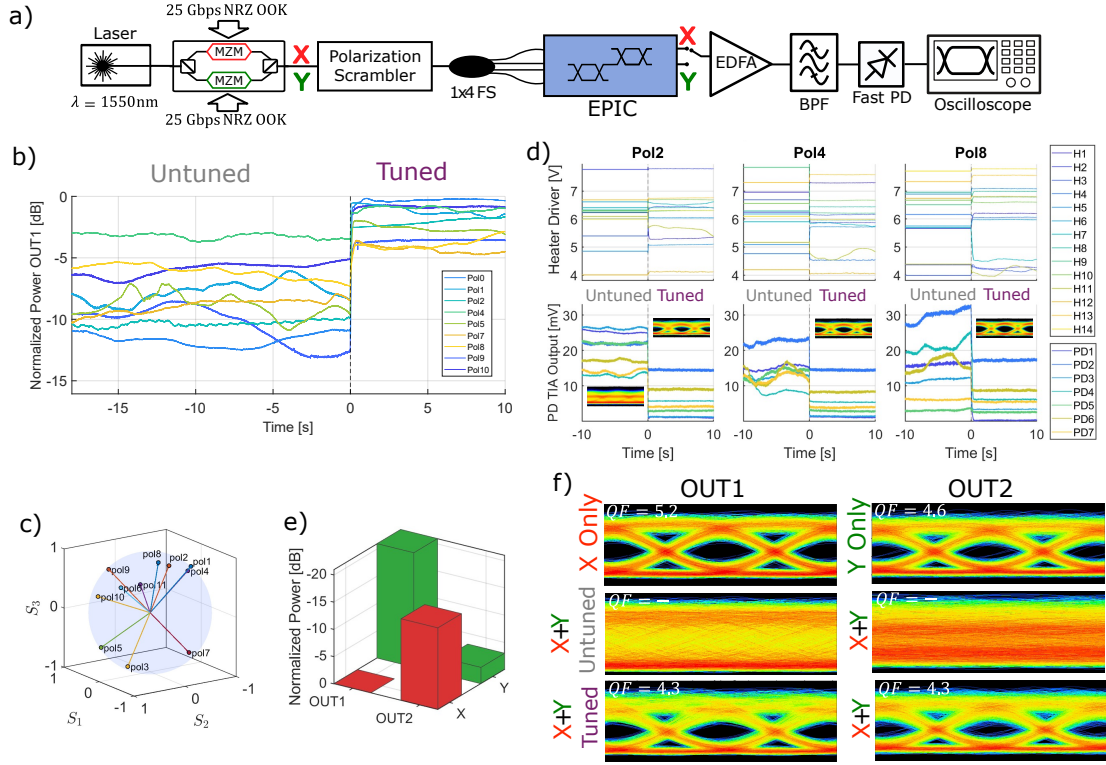


Fig. 3: (a) Schematic of the setup for the coherent combining and demultiplexing experiments; (b) plot of the normalized OUT1 power before and after tuning the MZI mesh for ten polarization-state cases; (c) Poincaré sphere showing the span of the generated polarization states; (d) bar chart showing the coupling and isolation between the two demultiplexed polarization states; (e) eye diagrams of the 25 Gbps signal for the cases of a single polarization state injected (top row), both polarization states injected with the mesh untuned (middle row), and the mesh tuned (bottom row).

configured to combine the input power toward their top outputs, the input signal is successfully stabilized at OUT1. Note that this stabilization is achieved within a range of less than 4 dB, which is attributed to the polarization-dependent loss of the custom-designed input edge couplers. Figure 3(c) shows the span of the generated polarization states on the Poincaré sphere.

The on-chip driver voltages and the amplified signals from the integrated PDs are shown in Fig. 3(d) for three representative cases. In the untuned case, the heater voltages are set to arbitrary values, and the PDs detect nonzero power depending on their position along the diagonal and on the input polarization, which fluctuates due to the phase variations at the chip input. After tuning the first MZI diagonal, the heater voltages are adjusted to compensate for these phase variations and continue to vary accordingly during operation. In this condition, the PD signals are minimized and maintained at their lowest values. The eye diagrams of the 25 Gbps signal are shown in the insets for both the untuned and tuned cases.

The demultiplexer experiment is then considered. In this case, both X and Y polarization states are generated and, after scrambling, are successfully retrieved at the two mesh outputs, OUT1 and OUT2. The mesh diagonals are continuously tuned, with the first diagonal assigned to polarization state X and the second diagonal

to state Y; however, this assignment remains arbitrary, since the mesh is transparent to the sorting operation. The bar chart in Fig. 3(e) shows the isolations obtained when only one state is injected and tuned at a time, either X or Y. High isolation is maintained by sequentially tuning the system for the “X Only” and “Y Only” states over time. The eye diagrams for these two cases are shown in the first row of Fig. 3(f).

The experiment is then repeated with both X and Y states generated simultaneously. In this case, stopping the tuning of the MZI mesh causes the eye diagrams to close, preventing signal retrieval, as shown in the second row of the figure. However, when the mesh is tuned for the X+Y state, the combined signals are successfully demultiplexed at the mesh outputs OUT1 and OUT2, as confirmed by the recovered eye diagrams.

Conclusions

The coherent combining of a single polarization state, as well as the demultiplexing of two orthogonal states after scrambling, are demonstrated. Using the same number of inputs, the approach can be extended beyond two diagonals to enable simultaneous polarization and mode demultiplexing in multimode fibers, with the structure acting as an interface between a multimode fiber and multiple single-mode fibers while performing all-optical MIMO processing.

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